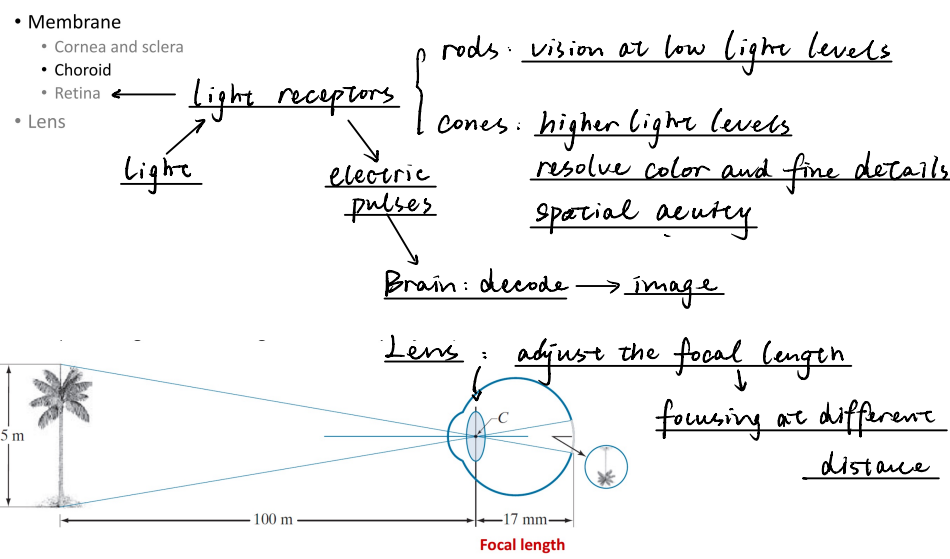


Fundamentals

Human Visual System

Eye = Sensor → Central nervous system

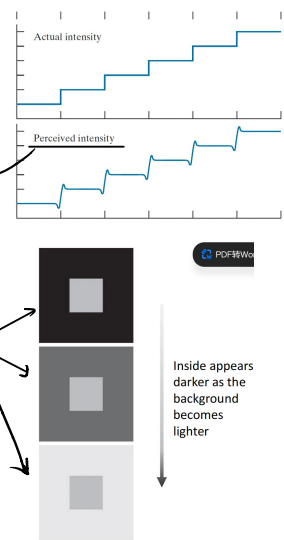


Brightness Discrimination

Overshoot/Undershoot around boundary

not just a simple function of intensity

Simultaneous Contrast



Optical illusion

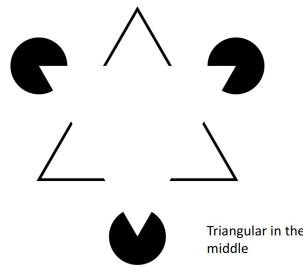
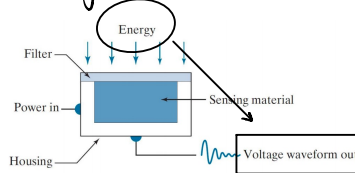


Image acquisition:

Source of energy → Sensor → Digital Image

reflection transmission
digitization

A single sensor element:



Sensor for 2D image:

① pros: inexpensive, cons: slow.

② In-Line Sensor Strip

A line of sensors:

③ Sensor array

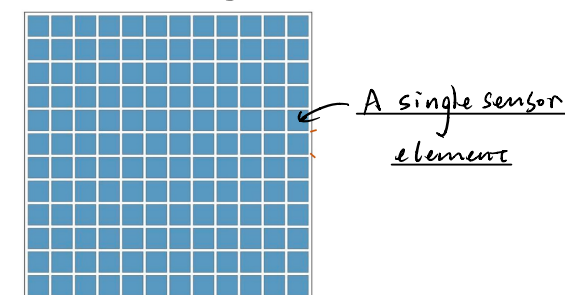
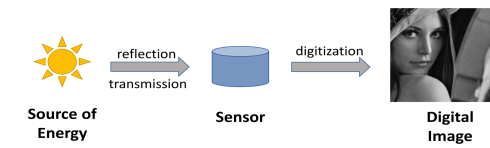


Image Formation Model

$f(x,y) = i(x,y) \cdot r(x,y)$
energy received by imaging system
illumination
reflection rate ($0 \leq r \leq 1$)



Digitalization

① Sampling ② Quantization
digitizing x-y coordinate values → record the value of energy
digitizing amplitude values → approximate to discrete value

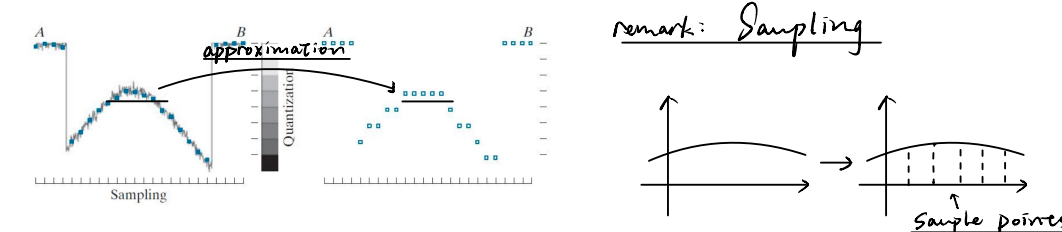
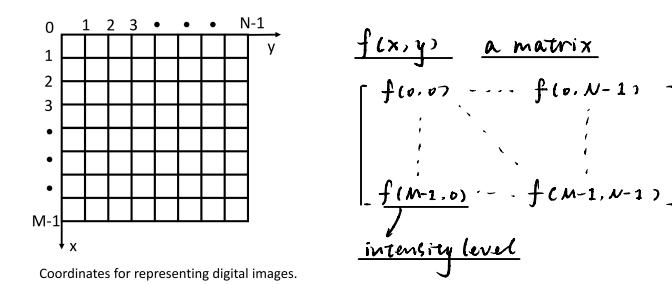


Image representation



Intensity levels: $L=2^k$

$L=2^k$: power of 2

for an image with size $M \times N$, # bits to store it: $M \times N \times k$ (smaller in reality due to compression)

intensity level ↑, intensity discrimination? (more sensitive to color change)

intensity level ↓ ⇒ false contouring: "wavy" approx "smooth" False Contouring

Spatial Resolution



- Smallest discernible details
- Line pairs per unit distance/ Dots (pixels) per unit distance
- Used with spatial units
- U.S.: dots per inch (dpi)

Intensity Resolution

8 bit → 256 bit

- Smallest discernible change in intensity level
- Can be described with no. of levels or no. of bits used

Pixel count: (often called) resolution

more pixel often means better image quality

But not always:

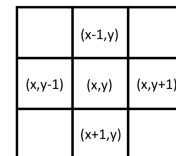


Basic relationships among pixels

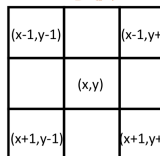
① Neighbours

$p=(x,y)$

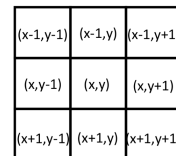
4-neighbors of p
 $N_4(p)$



Four diagonal neighbors of p
 $N_D(p)$



8-neighbors of p
 $N_8(p)$



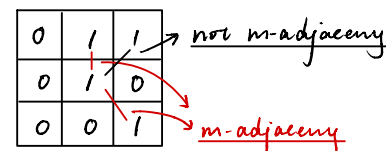
① Adjacency

4-adjacency: $q \in N_4(p)$

8-adjacency: $q \in N_8(p)$

m-adjacency: "mixed adjacency"

$q \in N_4(p)$ or $(q \in N_8(p) \text{ and } N_4(p) \cap N_8(p) = \emptyset)$



- 1) they are diagonal neighbor
- 2) they don't have any shared 4-neighbor

Connectivity

Digital Path: from p to q . Sequence of distinct pixels, $(x_i, y_i), (x_{i+1}, y_{i+1})$ are adjacent $1 \leq i \leq n$

① if $(x_0, y_0) = (x_n, y_n)$, it is a closed path

② if there is a path between p, q . We say p, q are Connected

Connected Component

For any p in S , the set of pixels that connected to it in S

Connected Set

S has only 1 Connected Component

Regions

R → a subset of pixels in the image.
→ A region
→ a connected set

Boundary of a region:

The set of pixels which has at least 1 background neighbors (4-, 8-, m-)

Outer boundary

Set of pixels → ① in background
→ ② has at least 1 neighbor from the region

Distance Measures

$p=(x,y)$ $q=(u,v)$

Euclidean distance:	$D_L(p,q) = [(x-u)^2 + (y-v)^2]^{1/2}$	2 2 2 2 1 2 2 1 0 1 2 2 1 2 2	Constant distance contour of D_L distance.
D_4 distance (city block distance):	$D_4(p,q) = x-u + y-v $		
D_8 distance (chessboard distance):	$D_8(p,q) = \max(x-u , y-v)$	2 2 2 2 2 2 1 1 1 2 2 1 0 1 2 2 1 1 1 2 2 2 2 2 2	Constant distance contour of D_8 distance.

Note the D_4 and D_8 distances between p and q are independent of any paths that might exist between these points because these distances involve only the coordinates of the points.

only depends on p, q

In the case of m -adjacency, however, the D_m distance between two points is defined as the shortest m -path between the points. In this case, the distance between two pixels will depend on the values of the pixels along the path, as well as the values of their neighbors.

path dependency